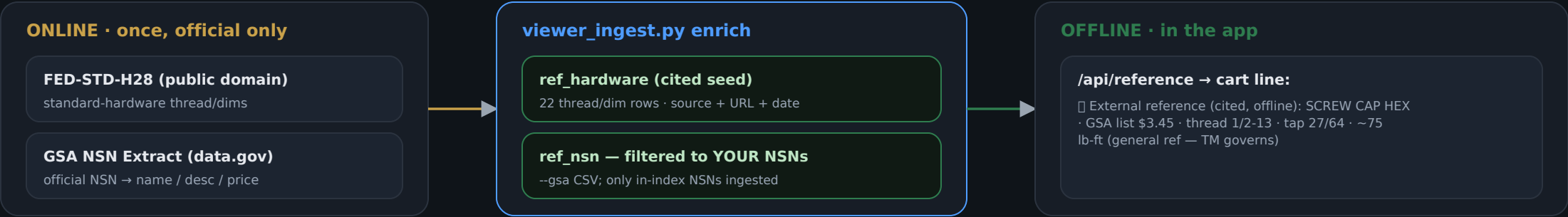


Online → offline reference enrichment — built

v0.20.0 · fetched ONCE from official public-domain sources · baked into separate, cited reference tables · engine stays offline. Additive (migration 0005).



PROVENANCE & SEPARATION (what keeps it grounded)

- Stored in SEPARATE tables (ref_hardware / ref_nsn) — never merged into manual citations, so external data can't pose as TM-sourced.
- Every row carries source + URL + fetch date, and is labeled '□ External reference (cited)' in the UI.
- Official / public-domain only (FED-STD-H28, GSA/data.gov). Third-party scrapers excluded by your choice.
- Torque shown is a GENERAL reference — the TM's stated torque governs. FEDLOG price/AAC/ARC still come from your AMDF (GSA shown as 'list', clearly labeled).
- NSN ingest is filtered to NSNs already in your index (targeted, relevant). Engine is offline after the one-time fetch.

WHAT IT UNLOCKS — and the remaining gap

Fills missing nomenclature on the request sheet, and gives cited dimensions for standard hardware (the Tier 2.5 key). For full parametric 3D you still need each part's specific SIZE — parsed from RPSTL text, the fuller NSN characteristics (FLIS) where available, or SME-confirmed; model only when size is certain (your decisions, recorded for Tier 2.5).